## Sequences

**1**

**a** $6$ **b** $4.5$ **c** $1$

**2**

**a** $6$ **b** $0.2$ **c** $7$ **d** $20$

**e** $6$ **f** $-56$

**3** $3$

**4**

**a**

**i** The sequence starts at $-1$ and goes up by $2$.

**ii** $9$

**b**

**i** The sequence starts at $64$ and each term is subsequently halved.

**ii** $1$

**c**

**i** The sequence starts at $2$ and each term goes up by $2$ in magnitude, while the sign of the terms alternate.

**ii** $14$

## Explicit rules

**5**

**a** No **b** Yes **c** No **d** Yes

**6**

**a** $3, 8, 13, 18, 23$ **b** $7, 10, 15, 22, 31$ **c** $0, 7, 18, 33, 52$

**7**

**a** $T_n = 1300 \times (1.025)^{n-1}$

**b** $T_n = 44 + 77\,(n-1)$

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